UNIFIED OBSERVABILITY FOR POSTGRESQL ON KUBERNETES

Logs, traces and metrics under one roof

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EDB POSTGRES AI

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SPEAKER

ABOUT ME

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APPRENTICE SPEAKER

I build Postgres observability for Kubernetes and traditional infrastructure.

Ten years from full-stack to backend and telemetry.

I lead teams and mentor engineers.

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THE SCENARIO

YOU'RE PAGED: CHECKOUT IS SLOW

ALERT · 03:12

CHECKOUT P95 LATENCY HIGH

Two PostgreSQL clusters behind it.

01 Which cluster?

02 Which query?

03 Why? What fixes it?

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THE PROBLEM

THREE TOOLS AND ONE BLIND SPOT

LOGS

Statements and errors

In a log aggregator.

METRICS

Instance health

In a time-series database.

TRACES

Stop at the application

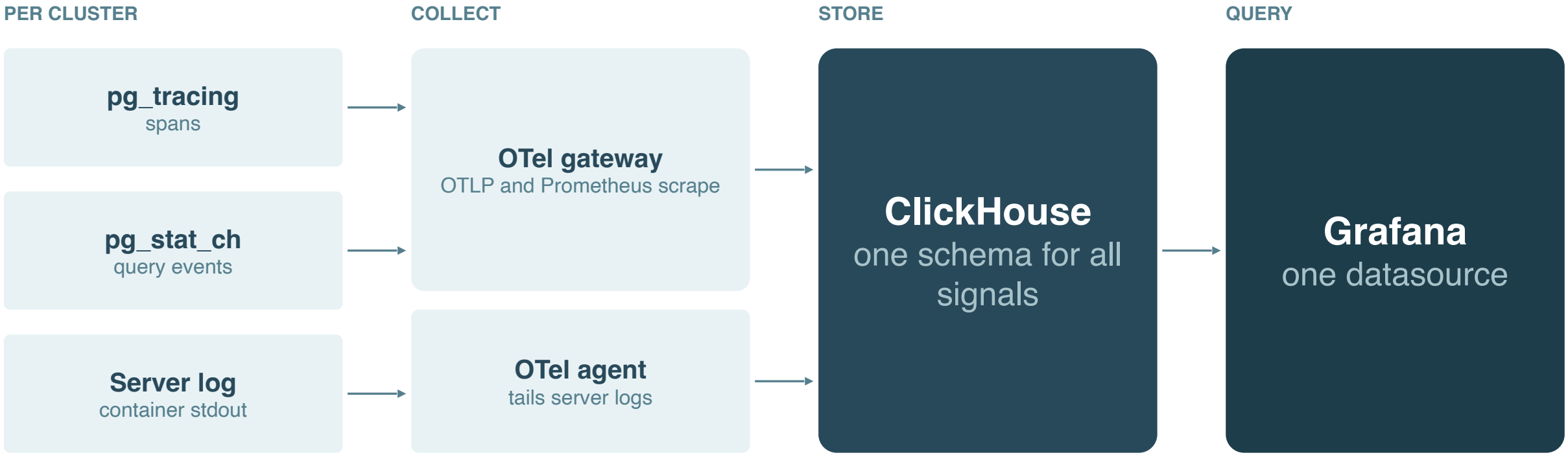
The span tree ends at the database client.

THE COST No path from a slow request to the SQL plan behind it.

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ARCHITECTURE

ONE PLATFORM WIRED ONCE



One namespace per cluster

Join key: query_id

Why ClickHouse: columnar, cheap to keep, fast to query.

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CLOUDNATIVEPG

PER CLUSTER: THREE SETTINGS

```
spec:
  imageCatalogRef:
    kind: ClusterImageCatalog
    name: o11y-postgres
    major: 18
  postgresql:
    shared_preload_libraries: [pg_tracing, pg_stat_ch]
    parameters:
      compute_query_id: "on"
      pg_tracing.otel_endpoint: http://otel-collector.o11y:4318/v1/traces
      pg_stat_ch.otel_endpoint: otel-collector.o11y:4317
```

01 **Load both extensions**

shared_preload_libraries

02 **Compute query ids**

The join key

03 **Point at the gateway**

OTLP for spans and events

ONE-TIME COST pg_tracing and pg_stat_ch live in one image shared by every cluster.

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LIVE DEMO: THE INCIDENT

Locate · Narrow · Diagnose · Fix · Verify

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WRAP-UP

TAKEAWAYS

01

Three pillars on one backend

One schema and one Grafana datasource for all three signals.

02

PostgreSQL inside the trace

pg_tracing extends the application trace into the plan and execution.

03

One join key

query_id links spans, events and logs across plan changes.

THE POINT From the page to the root cause in one Grafana view.

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NEXT STEPS

IMPROVEMENTS

01 **Extension image volumes**

Replace the custom image. Needs PG 18 and Kubernetes 1.35+.

02 **Log above a threshold**

Not every statement.

03 **Replicas and backups**

CNPG provides both.

04 **Retention**

Set a ClickHouse TTL.

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LET'S KEEP IN TOUCH

Open source (MIT). Scan, clone, break it.

DEMO REPOSITORY

github.com/gabrosys/postgres-k8s-unified-o11y

LINKEDIN

linkedin.com/in/gabriele-rossini

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BACKUP

QUERY_ID JOINS EVERY SIGNAL

```
otel_traces.SpanAttributes['query.query_id']
otel_logs.LogAttributes['query_id']   -- ServiceName = 'pg_stat_ch'
otel_logs.LogAttributes['query_id']   -- ServiceName = 'postgres-server-log'
```

Derived from the parse tree

Stable across plan changes.

Needs two settings

compute_query_id on and statement logging.

Metrics join by time and pod

They describe whole instances.

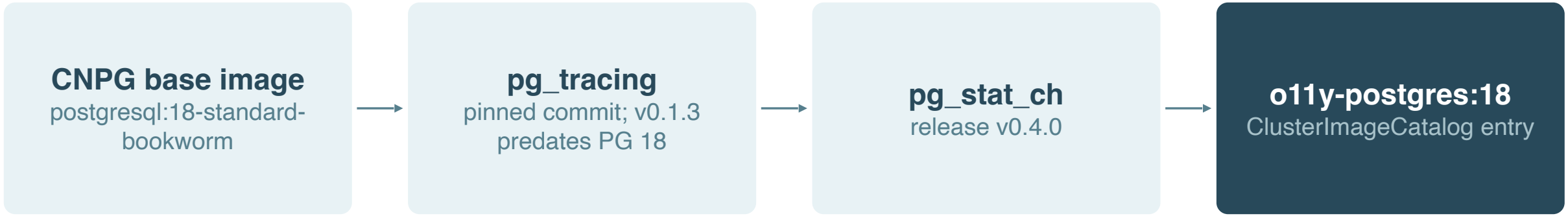
Traces and logs join by trace id

The traceparent is in the logged statement.

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BACKUP

THE EXTENSION IMAGE



Upstream -Werror is dropped so pg_tracing builds on PG 18.

pg_tracing links libcurl. The final stage installs it.

The image is loaded into KIND and referenced through the catalog.

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